CHAPTER 4

NETWORKS OF INFORMATION

A description of networks of information or data, with a particular focus on the World Wide Web and citation networks

THIS CHAPTER focuses on networks of information, networks consisting of items of data linked together in some way. Information networks are all, so far as we know, man-made, with perhaps the best known example being the World Wide Web, though many others exist and are worthy of study, particularly citation networks of various kinds. These and several other types of information networks are discussed in this chapter.

In addition, there are some networks which could be considered information networks but which also have social aspects to them. Examples include networks of email communications, networks on social-networking websites such as *Facebook* or *LinkedIn*, and networks of weblogs and online journals. These and similar examples were discussed in the previous chapter on social networks, in Section 3.4, but they would have fitted perfectly well in the present chapter also. The classification of networks as social networks, information networks, and so forth is a fuzzy one, and there are plenty of examples that, like these, straddle the boundaries.

4.1 THE WORLD WIDE WEB

Although by no means the first information network created, the World Wide Web is probably the example best known to most people and a good place to start our discussion in this chapter.

As described in Chapter 1, the Web is a network in which the vertices are web pages consisting of text, pictures, or other information and the edges are the hyperlinks that allow us to navigate from page to page. Since hyperlinks run in one direction only, the Web is a directed network. We can picture the network with an arrow on each edge indicating which way it runs. Some pairs of web pages are connected by hyperlinks running in both directions, which can be represented by two directed edges, one in each direction between the corresponding vertices. Figure 4.1 shows a picture of a small portion of the Web network, representing the connections between a set of web pages on a single website.

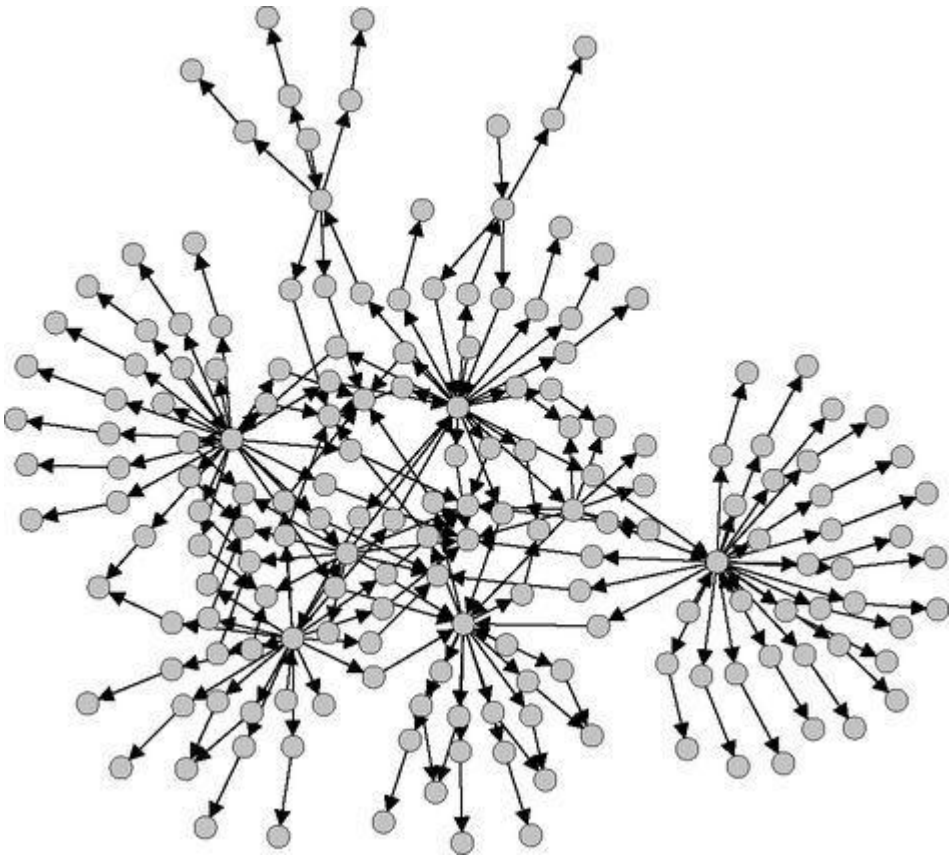


Figure 4.1: A network of pages on a corporate website. The vertices in this network represent pages on a website and the directed edges between them represent hyperlinks.

The World Wide Web was invented in the 1980s by scientists at the CERN high-energy physics laboratory in Geneva as a means of exchanging information among themselves and their coworkers, but it rapidly became clear that its potential was much greater [159]. At that time there were several similar ideas competing for dominance of the rapidly growing Internet, but the Web

won the battle, largely because its inventors decided to give away for free the software technologies on which it was based—the Hypertext Markup Language (HTML) used to specify the appearance of pages and the Hypertext Transport Protocol (HTTP) used to transmit pages over the Internet. The Web's extraordinary rise is now a familiar story and most of us use its facilities at least occasionally, and in some cases daily. A crude estimate of the number of pages on the Web puts that number at over 25 billion at the time of the writing of this book.[27] The network structure of the Web has only been studied in detail relatively recently however.

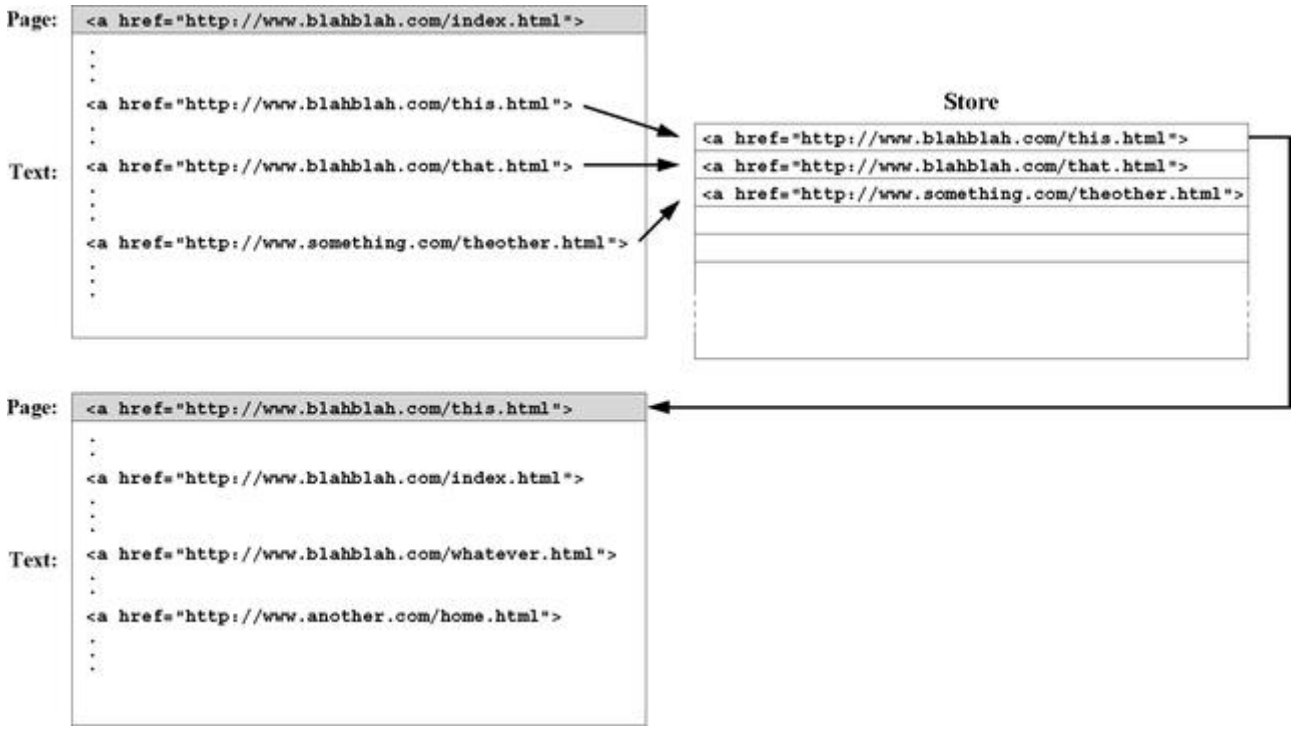


Figure 4.2: The operation of a web crawler. A web crawler iteratively downloads pages from the Web, starting from a given initial page. URLs are copied from the link tags in that initial page into a store. Once all links have been copied from the initial page, the crawler takes a URL from the store and downloads the corresponding page, then copies links from that, and so on.

Breadth-first search is discussed at length in Section 10.3.

The structure of the Web can be measured using a *crawler*, a computer program that automatically surfs the Web looking for pages. In its simplest form, the crawler performs a so-called breadth-first search on the Web network, as shown schematically in Fig. 4.2. One starts from any initial web page, downloads the text of that page over the Internet, and finds all the links in the text. Functionally, a link consists of an identifying "tag"—a short piece of text marking the link as a link—and a *Uniform Resource Locator* or URL, a standardized computer address that says how and where the linked web page can be found. By scanning for the tags and then copying the adjacent URLs a web crawler can rapidly extract URLs for all the links on a web page, storing them in memory or on a disk drive. When it is done with the current page, it removes one of the URLs from its store, uses it to locate a new page on the Web, and downloads the text of that page, and so the process repeats. If at any point the crawler encounters a URL that is the same as one already in its store, then that URL is ignored and not added to the store again, to avoid duplicate entries. Only URLs that are different from all those seen before are added to the store.

By repeating the process of downloading and URL extraction for a suitably long period of time

one can find a significant portion of the pages on the entire Web. In practice, however, no web crawler actually finds all the pages on the Web. There are a number of reasons for this. First, some websites forbid crawlers to crawl their pages. Websites can place a file called robots . txt in their root directory that specifies which files, if any, crawlers can look at and may optionally specify that some crawlers are allowed to look at files while others are not. Compliance with the restrictions specified in a robots . txt file is voluntary, but in practice many crawlers do comply.

Second, many pages on the Web are dynamically generated: they are created on the fly by special software using, for instance, data from a database. Many corporate websites, as well as the web pages generated by search engines or directory services, fall into this category. The number of possible web pages that can be displayed as a result of a search using the *Google* search engine, for example, is so large as to be effectively infinite; it would not be possible (or sensible) for a crawler to crawl all of these pages. The crawler therefore has to make some choice about what counts as a web page and what does not. One choice would be to restrict oneself to static web pages—ones that are not generated on the fly. But it's not always simple to tell which pages are static, and besides, much useful information resides in the dynamic pages. In practice, the decisions made by crawlers about which pages to include tend to be fairly arbitrary, and it is not easy to guess which pages will be included in a crawl and which will not. But one can say with certainty that many will not and in this sense the crawl is always incomplete.

However, perhaps the most important reason why web crawls do not reach all the pages on the Web is that the network structure of the Web does not allow it. Since the Web is a directed network, not all pages are reachable from a given starting point. In particular, it is clear that pages that have no incoming hyperlinks—pages that no one links to—can never be found by a crawler that blindly follows links. Taking that idea one step further, it is also the case that a page will never be found if it is only linked to by pages that themselves have no incoming links. And so forth. In fact, the Web, and directed networks in general, have a special "component" structure, which we will examine in detail in Section 6.11.1, and most crawlers only find one part of that structure, the "giant out-component." In the case of the World Wide Web the giant out-component constitutes only about a half of all web pages and the other half of the Web is unreachable.[28]

Although we are interested in web crawlers as a tool for probing the structure of the Web so that we can study its network properties, this is not their main purpose. The primary use of web crawlers is to construct directories of web pages for search purposes. Web search engines such as *Google* indulge in web crawling on a massive scale to find web pages, parse their content, and construct indexes of the words and pictures they contain that can later be searched offline by fast database engines to find pages of interest to searchers. Because their primary interest is in indexing, rather than in reconstructing the network structure of the Web, search engine companies don't have any particular reason to take a good statistical sample of the Web and in network terms their crawls are probably quite biased. Still, many of them have graciously made their data available to academic researchers interested in web structure, and the data are good enough to give us a rough picture of what is going on. We will study a variety of features of the Web network in subsequent chapters.

> Web searching, which itself raises some interesting network questions, is discussed in Section 19.1.

It isn't entirely necessary that we rely on search engine companies or other web enterprises for data on the structure of the Web. One can also perform one's own web crawls. There are a number of excellent web crawlers available for free, including *wget*, *Nutch*, *GRUB*, *Larbin*, *WebSPHINX*, and *ht://Dig*. While most of us don't have the time and network bandwidth to crawl billions of web pages, these programs can be useful for crawling single websites, and much useful insight and information can be acquired by doing so.

4.2 CITATION NETWORKS

A less well-known but much older information network is the network of citations between academic papers. Most papers refer to one or more other previous papers, usually in a bibliography at the end of the paper, and one can construct a network in which the vertices are papers and there is a directed edge from paper A to paper B if A cites B in its bibliography. There are many reasons why one paper might cite another—to point out information that may be useful to the reader, to give credit for prior work, to indicate influences on current work, or to disagree with the content of a paper. In general, however, if one paper cites another it is usually an indication that the contents of the earlier paper are relevant in some way to those of the later one, and hence citation networks are networks of relatedness of subject matter.

Quantitative studies of citation networks go back to the 1960s. The earliest seems to be the 1965 study by Price [274] (which is also the earliest study we know of to find a "power-law degree distribution," of which we talk in detail in Section 8.4). Studies such as this usually fall within the field formerly known as "library science" but now more often called "information science." The branch of information science dealing specifically with the statistical study of publications and citations is called *bibliometrics*.

The most common way to assemble citation data is to do it by hand, simply typing in all the entries in the bibliographies of papers to create a database that can then be used to assemble the network. In the 1960s when Price carried out his study, such databases were just starting to be created and he made use of an early version of what would later become the Science Citation Index. The Science Citation Index (along with its sister publications, the Social Science Citation Index and the Arts and Humanities Citation Index) is now one of the primary and most widely used sources of citation data. Another database, Scopus, provides a competing but largely similar service. Both are hand-maintained by professional staff and their coverage of the literature is reasonably complete and accurate, although the data are also quite expensive to purchase. Still, if one has the money, creating a citation network is only a matter of deciding which papers one wishes to include, using one of the databases to find the citations between those papers, and adding the appropriate directed edges to the network until it is complete.

See Section 4.1 for a discussion of web crawlers.

More recently, automated citation indexing by computer has started to become more common. For instance, the website *Citeseer*, maintained by Pennsylvania State University, performs citation indexing of papers in computer science and information science by crawling the Web to find freely available manuscripts of papers in electronic form, and then searching through those manuscripts to identify citations to other papers. This is a somewhat hit-ormiss operation because many papers are not on the Web or are not freely available, citations in papers have a wide variety of different formats and may include errors, and the same paper may exist in more than one place on the Web as well as in journals or books, and possibly in more than one different version. Nonetheless, enough progress has been made for Citeseer to become a useful tool in the computer science community. Other automatic citation indexing projects include *Citebase*, which indexes physics papers, and *Google Scholar*.

As with web crawls, the primary purpose of citation indexes is not to allow us to study the network structure of citation. Citation indexes are primarily research tools that allow researchers to discover by whom a paper has been cited, and hence to find research related to a topic of interest. Nonetheless, data from citation indices have been widely used to reconstruct the underlying networks and study their properties.

Citation networks are in many ways similar to the World Wide Web. The vertices of the network hold information in the form of text and pictures, just as web pages do, and the links from one paper to another play a role similar to hyperlinks on web pages, alerting the reader when information relevant to the topic of one paper can be found in another.[29] Papers with many citations are often more influential and widely read than those with few, just as is the case with web pages, and one can "surf " the citation network by following a succession of citations from paper to paper just as computer users surf the Web.

Acyclic networks are discussed further in Section 6.4.2.

There is, however, at least one important difference between a citation network and the Web: a citation network is *acyclic*, while the Web is not. An acyclic network is one in which there are no closed loops of directed edges. On the World Wide Web, it is entirely possible to follow a succession of hyperlinks and end up back at the page you started at. Indeed this happens often. On a citation network, by contrast, it is essentially impossible. The reason is that in order to cite a paper, that paper must already have been written. One cannot cite a paper that doesn't exist yet. Thus all the directed edges in a citation network point backward in time, from newer papers to older ones. If we follow a path of such edges from paper to paper, we will therefore find ourselves going backward in time, but there is no way to go forward again, so we cannot close the loop and return to where we started.[30]

See Fig. 6.3 for an illustration of a small acyclic network.

Citation networks have some surprising statistics. About 47% of all papers in the Science Citation Index have never been cited at all. Of the remainder, 9% have one citation, 6% have two, and it goes down quickly after that. Only 21% of all papers have 10 or more citations, and just 1% have 100 or more. These figures are a consequence of the power-law degree distribution of the network mentioned above and discussed more in Section 8.4.

The most highly cited paper in the Science Citation Index is a paper by Lowry *et al.* [202], which has been cited more than a quarter of a million times.[31] Like most very highly cited papers, it is a methodological paper in molecular biology.

Citation networks of the type described so far are the simplest but not the only possible network representation of citation patterns. An alternative and widely studied representation is the *cocitation network*. Two papers are said to be cocited if they are both cited by the same third paper. Cocitation is often taken as an indicator that papers deal with related topics and there is good evidence that this is a reasonable assumption in many cases.

A cocitation network is a network in which the vertices represent papers and the edges represent cocitation of pairs of papers. By contrast with ordinary citation networks, the edges in a cocitation network are normally considered undirected, since cocitation is a symmetric relationship. One can also define a strength for the cocitation between two papers as the number of other papers that cite both and one can create weighted cocitation networks in which the strengths of the edges are equal to this cocitation strength.

Another related concept, although one that is less often used, is *bibliographic coupling*. Two papers are said to be bibliographically coupled if they *cite* the same other papers (rather than being cited by the same papers). Bibliographic coupling, like cocitation, can be taken as an indicator that papers deal with related material and one can define a strength or weight of coupling by the number of common citations between two papers. From the bibliographic coupling figures one can then assemble a *bibliographic coupling network*, either weighted or not, in which the vertices are papers and the undirected edges indicate bibliographic coupling.

Cocitation and bibliographic coupling are discussed in more detail in Section 6.4.1.

4.2.1 PATENT AND LEGAL CITATIONS

Our discussions of citation networks have so far focused on citations between academic papers, but there are other types of citation also. Two of particular importance are citations between patents and between legal opinions.

Patents are temporary grants of ownership for inventions, which give their holders the right to take legal action against others who attempt to profit without permission from the protected inventions. They are typically issued to inventors—either individuals or corporations—by national governments after a review process to determine whether the invention in question is original and has not previously been invented by someone else. In applying for a patent, an inventor must describe his or her invention in sufficient detail to make adequate review possible and present the case that the invention is worthy of patent protection. A part of this case typically involves detailing the relationship between the invention and previously patented inventions, and in doing so the inventor will usually cite one or more previous patents. Citations may highlight dependencies between technologies, such as one invention depending for its operation on another, but more often patent citations are "defensive," meaning that the inventor cites the patent for a previous technology and then presents an argument for why the new technology is sufficiently different from the old one to merit its own patent. Governments, in the process of examining patent applications, will routinely consider their similarity to previous inventions, and defensive citations are one way in which an inventor can fend off in advance possible objections that might be raised. Typically there are a number of rounds of communication, back and forth between the government patent examiner and the inventor, before a patent application is finally accepted or rejected. During this process extra citations are often added to the application, either by the inventor or by the examiner, to document the further points discussed in their communications.

If and when a patent is finally granted, it is published, citations and all, so that the public may know which technologies have patent protection. These published patents provide a source of citation data that we can use to construct networks similar to the networks for citations between papers. In these networks the vertices are patents, each identified by a unique patent number, and the directed edges between them are citations of one patent by another. Like academic citation networks, patent networks are mostly acyclic, with edges running from more recent patents to older ones, although short loops can arise in the network in the not uncommon case that an inventor simultaneously patents a number of mutually dependent technologies. The structure of patent networks reflects the organization of human technology in much the same way that academic citations reflect the structure of research knowledge. Patent networks have been studied far less than academic citation networks, but studies have been growing in the last few years with the appearance of high-quality data sets, particularly for US patents [161], and there are a number of important technological and legal questions, for instance concerning antitrust policy, that can be addressed by examining their structure [69].

Another class of citation network that has begun to attract attention in recent years is that of legal citation networks. In countries where law cases can be heard by judges rather than juries, such as civil cases or appeals in Europe or the US, a judge will frequently issue an "opinion" after deciding a case, a narrative essay explaining his or her reasoning and conclusions. It is common practice in writing such an opinion to cite previous opinions issued in other cases in order to establish precedent, or occasionally to argue against it. Thus, like academic papers and patents, legal opinions form a citation network, with opinions being the vertices and citations being the directed edges. Again the network is approximately acyclic, as with the other networks in this section. The legal profession has long maintained indexes of citations between opinions for use by lawyers, judges, scholars, and others, and in recent years those indexes have made the jump to electronic form and are now available online. In the United States, for instance, two commercial

services, LexisNexis and Westlaw,[32] provide thorough and detailed data on legal opinions and their citations via online services. In the last few years a number of studies have been published of the structure of legal citation networks using data derived from these services [125, 126, 194].

In principle it would be possible also to construct networks of cocitation or bibliographic coupling between either patents or legal opinions, but the author is not aware of any studies yet published of such networks.

4.3 OTHER INFORMATION NETWORKS

There are many other networks of information, although none have received the same level of study as the Web and citation networks. In the remainder of this chapter we briefly discuss a few examples of other networks.

4.3.1 PEER-TO-PEER NETWORKS

Peer-to-peer (P2P) file-sharing networks have become popular and widespread in the last decade or so. A *peer-to-peer network* is a network in which the nodes are computers containing information in the form, usually, of discrete files, and the edges between them are virtual links established for the purpose of sharing the contents of those files. The links exist only in software—they indicate only the intention of one computer to communicate with another should the need arise.

Peer-to-peer networks are typically used as a vehicle for distributed databases, particularly for the storage and distribution, often illegally, of music and movies, although there are substantial legal uses as well, such as local sharing of files on corporate networks or the distribution of open-source software. (The network of router-to-router communications using the Border Gateway Protocol described in Section 2.1 is another less obvious example of a legitimate and useful peer-to-peer network.)

The point of a peer-to-peer network is that data is transferred directly between computers belonging to two end users of the network, two "peers." This contrasts with the more common server-client model, such as that used by the World Wide Web, in which central server computers supply requested data to a large number of client machines. The peer-to-peer model is favored particularly for illicit sharing of copyrighted material because the owners of a centralized server can easily be obliged to turn off the server by legal or lawenforcement action, but such actions are much more difficult when no central server exists.

On most peer-to-peer networks each computer is home to some information, but no computer has all the information in the network. If the user of a computer requires information stored on another computer, that information can be transmitted simply and directly over the Internet or over a local area network. This is a peer-to-peer transfer, but no special infrastructure is necessary to accomplish it—standard Internet protocols are perfectly adequate to the task. Things get interesting, however, when one wants to *find* which other computer has the desired information. One way to do that is to have a central server containing none of the information but just an index of which information is on which computers. Such a system was employed by the early file-sharing network *Napster*, but the central index server is, once again, susceptible to legal and other challenges, and such challenges were in the end responsible for shutting Napster down.[33]

To avoid this problem, developers have turned to distributed schemes for searching and this is where network concepts come into play. An illustrative example of a peer-to-peer system with distributed search is the *Gnutella* network, which underlies a number of popular file-sharing programs including *LimeWire* and the now-defunct *Morpheus*. In the simplest incarnation of this system (more sophisticated ones are in use now) computers form links to some number of their peers in such a way that all the computers form a connected network. Again, a link here is purely a software construct—a computer 's network neighbors in the peer-to-peer sense are merely those others with which it intends to communicate when the need arises.

When a user instructs his or her computer to search the network for a specific file the computer sends out a message to its network neighbors asking whether they have that file. If they do, they arrange to transmit it back to the first computer. If they do not, they pass the message on to *their* neighbors, and so forth until the file is found. As pointed out in Section 19.2, where we discuss search strategies on peer-to-peer networks at some length, this algorithm works, but only on relatively small networks. Since it requires messages to be passed between many computers for each individual search, the algorithm does not scale well as the network becomes large, the volume of network traffic eventually swamping the available data bandwidth. To get around this problem, modern peer-to-peer networks, including recent versions of Gnutella, employ a two-tiered network topology of nodes and "supernodes," in which searches are performed only among the supernodes

and ordinary nodes contact them directly to request searches be performed. More details are given in Section 19.2.

So what is the structure of a peer-to-peer network like? In many cases, unfortunately, not a lot is known since the software is proprietary and its owners are reluctant to share operational details. The Gnutella system is more promising, being so-called open-source software, meaning that the original computer code for the software and the specification of the protocols it uses are freely available. By exploiting certain details of these protocols, particularly the ability for computers in the Gnutella network to “ping” one another (i.e., ask each other to identify themselves), a number of authors have been able to discover structures for Gnutella networks [282, 308]. The networks appear to have approximately power-law degree distributions and it has been suggested that this property could be exploited to improve search performance [6].

4.3.2 RECOMMENDER NETWORKS

A type of information network important for technology and commerce is the *recommender network*. Recommender networks represent people's preferences for things, such as for certain products sold by a retailer. Online merchants, for instance, usually keep records of which customers bought which products and sometimes ask them whether they liked the products or not. Many large supermarket chains record the purchases made by each of their regular customers (usually identified by a small card with a barcode on it that is scanned when purchases are made) and so can work out which products each customer buys frequently.

The fundamental representation of a recommender network is as a "bipartite network," a network with two types of vertex, one representing the products or other items and the other representing the people, with edges connecting people to the items they buy or like. One can also add strengths or weights to the edges to indicate, for instance, how often a person has bought an item or how much he or she likes it, or the strengths could be made negative to indicate dislikes.

> We encountered bipartite networks previously in Section 3.5 and will study them further in Section 6.6.

Recommender networks have been studied for many types of goods and products, including books, music, films, and others. The primary commercial interest in recommender networks arises from their use in *collaborative filtering systems*, also sometimes called *recommender systems*, which are computer algorithms that attempt to guess items that people will like by comparing a person's known preferences with those of other people. If person A likes many of the same things as persons B, C, and D, and if persons B, C, and D all like some further item that A has never expressed an opinion about, then maybe (the theory goes) A would like that item too. A wide variety of computer algorithms have been developed for extracting conclusions of this type from recommender networks and are used extensively by retailers to suggest possible purchases to their customers, in the hope of drumming up business. The website of the online bookseller *Amazon.com*, for instance, has a feature that lists recommended book titles to customers based on their previously expressed preferences and purchases. And many supermarkets now print out discount coupons after customers have completed their purchases, coupons for products that the customer has not bought in the past but might be interested to try.

Research on recommender networks has in the past focused mainly on the development of new collaborative filtering algorithms, but it is reasonable to suppose that the success of these algorithms should depend to some extent on the structure of the recommender network itself, and there is therefore good reason to also study that structure. A few such studies have been published in the scientific literature [63, 147], but there is clearly room for further work.

4.3.3 KEYWORD INDEXES

Another type of information network, also bipartite in form, is the *keyword index*. Consider, for instance, a set of documents containing information on various topics. One can construct an index to that set so that one can look up words in that index and the index will list important occurrences of those words in the documents. Such indexes have historically appeared, of course, in books, as guides to their content, but more recently indexes have regularly been constructed as guides to other information collections, including sets of academic papers and the World Wide Web. The index constructed by a web search engine, as discussed in Section 4.1, is a good example; it consists, at a minimum, of a list of words or phrases, with each word or phrase accompanied by a list of the web pages on which it occurs.

Such indexes can be represented as a bipartite network in which one of the two types of vertex represents words in the index and the other represents documents or pages. Then one places an edge between each word and the documents in which it occurs. Although such networks can be constructed for, amongst other things, the Web or collections of academic papers, they should not be confused with the networks of web links or citations discussed earlier in this chapter. Those are also networks of web pages and documents, but they are different from a keyword index. Those networks were networks of direct links between documents. An index is a network of links between index entries and the documents they point to.

Indexes are of practical importance as a method for searching large bodies of information. Web search engines, for example, rely heavily on them to quickly find web pages that correspond to a particular query. However, indexes also have other, more sophisticated applications. They are used, for example, as a basis for techniques that attempt to find documents or pages that are similar to one another. If one has a keyword index to a set of documents and finds that two documents share a lot of the same keywords, it may be an indication that the two cover similar topics. A variety of computer algorithms for spotting such connections have been developed, typically making use of ideas very similar to those used in the recommender systems discussed above—the problem of finding documents with similar keywords is in many ways similar to the problem of finding buyers who like similar products.

The identification of similar documents can be useful, for example, when searching through a body of knowledge. In a standard index search, one typically types in a set of keywords and gets back a list of documents containing those words. Search engines that can tell when documents are similar to each other may be able to respond more usefully to such queries because they can return documents that do not in fact contain the keywords entered, but which are similar to documents that do. In cases where a single concept is called by more than one name, this may be a very effective strategy for finding all of the relevant documents.

In the context of document retrieval, the classic method for determining document similarity and performing generalized searches of this type is *latent semantic indexing*, which is based on the application of the matrix technique known as singular value decomposition to the bipartite network of keywords and documents. The interested reader can find a discussion of latent semantic indexing in Ref. [193].

As with recommender systems, it is reasonable to suppose that the success of methods for finding similar documents or improving searches using similarity information depends on the structure of the bipartite keyword/document network, and hence that studies of that structure could generate useful insights. There has been relatively little interest in the problem within the network community so far and again there is plenty of room for future work.